

LU & Cholesky Decomposition

**A high-performance mathematical guide to matrix split principles
and symmetric positive-definite systems.**

Part I: LU Decomposition

The standard for general square matrix factorization

The Split Principle

Fundamental Factorization

LU decomposition breaks down a square matrix into the product of two triangular matrices: a Lower (L) and an Upper (U).

$$A = L \times U$$

The Components

L (Lower): Lower-triangular matrix, often with 1s along the diagonal in Doolittle's method.

U (Upper): Upper-triangular matrix containing the result of Gaussian elimination.

$$\begin{bmatrix} A_{00} & A_{01} & A_{02} \\ A_{10} & A_{11} & A_{12} \\ A_{20} & A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ L_{10} & 1 & 0 \\ L_{20} & L_{21} & 1 \end{bmatrix} \begin{bmatrix} U_{00} & U_{01} & U_{02} \\ 0 & U_{11} & U_{12} \\ 0 & 0 & U_{22} \end{bmatrix}$$

Lower Triangular

Upper Triangular

| Solving Linear Systems

Instead of expensive matrix inversion, a linear system $Ax = b$ is resolved elegantly using two triangular substitutions:

- Solve $Ly = b$ for y via forward substitution.
- Solve $Ux = y$ for x via backward substitution.

This approach is significantly more stable and computationally efficient for multiple b vectors.

The diagram illustrates the LU decomposition of a 3x3 matrix A into a lower triangular matrix L and an upper triangular matrix U . The matrix A is shown on the left with elements A_{00}, A_{01}, A_{02} in the first row, A_{10}, A_{11}, A_{12} in the second row, and A_{20}, A_{21}, A_{22} in the third row. This is followed by an equals sign and two matrices. The first matrix is the lower triangular matrix L , which has 1s on the diagonal and elements L_{10}, L_{20}, L_{21} in the lower triangle. An arrow points from the label "Lower Triangular" to this matrix. The second matrix is the upper triangular matrix U , which has elements U_{00}, U_{01}, U_{02} in the first row, $0, U_{11}, U_{12}$ in the second row, and $0, 0, U_{22}$ in the third row. An arrow points from the label "Upper Triangular" to this matrix.

$$\begin{bmatrix} A_{00} & A_{01} & A_{02} \\ A_{10} & A_{11} & A_{12} \\ A_{20} & A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ L_{10} & 1 & 0 \\ L_{20} & L_{21} & 1 \end{bmatrix} \begin{bmatrix} U_{00} & U_{01} & U_{02} \\ 0 & U_{11} & U_{12} \\ 0 & 0 & U_{22} \end{bmatrix}$$

Numerical Example (3x3)

Row	Target Matrix A	Lower L (Unit Diag)	Upper U
Row 1	[1, 2, 4]	[1, 0, 0]	[1, 2, 4]
Row 2	[3, 8, 14]	[3, 1, 0]	[0, 2, 2]
Row 3	[2, 6, 13]	[2, 1, 1]	[0, 0, 3]

Part II: Cholesky Decomposition

Optimized for Symmetric Positive-Definite (SPD) Matrices

| Efficiency Gains

50
%

Fewer Operations

Cholesky requires exactly half the floating-point operations (FLOPs) of general LU decomposition by exploiting matrix symmetry.

It also requires half the memory, as only the single factor L needs to be stored.

| Strict Structural Preconditions

- ✓ **Symmetry:** The matrix must be identical to its transpose. Off-diagonal elements must match perfectly.
- ✓ **Positive-Definiteness:** For any non-zero vector x , the quadratic form $x^T A x$ must be strictly positive.
- ✓ **Real Values:** All eigenvalues must be strictly positive to avoid imaginary components during square root calculations.
- ✓ **Non-Singularity:** The matrix must be full rank. Zero pivots lead to immediate computational failure.

Mathematical Relationship

When matrix A is symmetric, LU simplifies to LDU.

Distributing diagonal scaling yields the Cholesky factor.

$$A = L \times L^T$$

This creates a single lower triangular factor multiplied by its own transpose, acting as a "square root" for the matrix.

| Cholesky Calculation Flow

Diagonal Elements

Calculated using square roots of residuals:

$$L_{jj} = \sqrt{A_{jj} - \sum_{k=1}^{j-1} L_{jk}^2}$$

Off-Diagonal Elements

Calculated by dividing the residual by the current pivot:

$$L_{ij} = \frac{1}{L_{jj}} A_{ij} - \sum_{k=1}^{j-1} L_{ik} L_{jk}$$